
AWS IoT Core – Overview

What is AWS IoT Core?

AWS IoT Core is a **managed cloud platform** that allows **connected devices** (sensors, appliances, vehicles, etc.) to interact securely with **cloud applications** and **other devices**. It handles **device connectivity, messaging, processing, and integration** with other AWS services.

Key Components of AWS IoT Core

Component	Description
 Devices	Physical IoT devices (sensors, cameras, etc.)
 Device Gateway	Entry point for MQTT, HTTPS, and WebSocket connections
 Message Broker	Publishes/subscribes messages using MQTT protocol
 Rules Engine	Allows filtering, transforming, and routing of device messages to AWS services
 Device Registry	Stores metadata (name, type) for devices
 Device Shadow	JSON document that stores device's last reported state
 Authentication & Authorization	Based on X.509 certs, IAM policies, and AWS IoT policies

How It Works

1. A device connects securely to **AWS IoT Core** using **MQTT/HTTPS/WebSockets**
2. The **message broker** handles communication with other devices and services

3. Messages are filtered via the **Rules Engine** and forwarded to:

- **Lambda**
- **DynamoDB**
- **Kinesis**
- **S3**
- **SNS/SQS**
- **CloudWatch**



Supported Protocols

Protocol	Use Case
MQTT	Lightweight, ideal for IoT
MQTT over WebSockets	Browser-based apps
HTTPS	Secure RESTful communication
LoRaWAN	For low-power, long-range devices (via AWS IoT Core for LoRaWAN)



Security Mechanisms

Layer	Security Feature
Transport	TLS Encryption
Identity	X.509 Certificates (per device)
Authorization	IAM + IoT Policies
Monitoring	AWS CloudWatch + AWS IoT Device Defender (advanced)

Device Shadow Example

```
{
  "state": {
    "desired": {
      "temperature": 22
    },
    "reported": {
      "temperature": 20
    }
  }
}
```

💡 Allows apps to **interact with devices even when offline**, and lets devices **synchronize with desired states**.

Integration with AWS Services

AWS Service	Use Case
Lambda	Run code in response to IoT messages
DynamoDB	Store time-series data
S3	Store sensor files or logs
Kinesis	Stream data for real-time analytics
SageMaker	Build predictive models from device data
SNS/SQS	Send alerts or messages
EventBridge	Trigger workflows or automation

Pricing Overview

Feature	Cost
Connectivity (MQTT/HTTP)	\$1.00 per million messages
Rules Engine	\$0.15 per million rules evaluations

Device Shadow \$1.25 per million operations

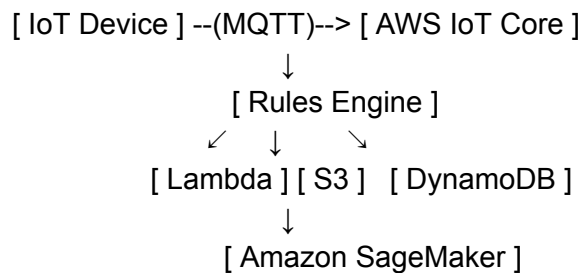
LoRaWAN Gateway & Devices Pay-as-you-go model

⚠️ AWS Free Tier: 250,000 free messages per month for 12 months

Use Cases

Use Case	Description
 Smart Home	Control lights, locks, appliances via cloud
 Industrial IoT	Monitor and control factory equipment
 Healthcare	Remote patient monitoring with secure data
 Automotive	Fleet management and telemetry
 Agriculture	Crop/soil monitoring with sensors
 Predictive Maintenance	Monitor equipment health via telemetry

Architecture Example



Advantages

- ✓ Scalable, serverless IoT platform
- ✓ Multiple protocol support
- ✓ Deep integration with AWS services

- ✓ Real-time messaging with low latency
 - ✓ Built-in security with X.509, IAM, and TLS
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! Challenges / Considerations

- Requires **careful policy management** for security
 - High throughput may require **IoT Device Defender** and **IoT Analytics**
 - Needs **device-side certificate management**
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